

REVIEW ARTICLE



Hydrogen sulfide poisoning in forensic pathology and toxicology: mechanism and metabolites quantification analysis

Jiaxin Zhang*, Longda Ma*, Yu Liu, Xin Tong and Yiwu Zhou

Department of Forensic Medicine, Tongji Medical College of Huazhong University of Science and Technology, Wuhan, China

ABSTRACT

Historically, hydrogen sulfide (H₂S) poisoning has extremely high and irreparable mortality. Currently, the identification of H₂S poisoning needs to combine with the case scene analysis in forensic medicine. The anatomy of the deceased seldom had obvious features. There are also a few reports about H₂S poisoning in detail. As a result, we give a comprehensive analysis of the related knowledge on the forensic aspect of H₂S poisoning. Furthermore, we provide the analytical methods of H₂S and its metabolite—which may assist in H₂S poisoning identification.

Abbreviations: 3MST: 3-mercaptopyruvate sulfurtransferase; AgNPs: silver nanoparticles; BP: blood pressure; CAT: cysteine aminotransferase; CBS: cystathionine-β-synthase; CCO: cytochrome C oxidase; CMQT: 2-chloro-1-methylquinolinium tetrafluoroborate; CO: carbon monoxide; CSE: cystathionine γ-lyase; DAO: D-amino acid oxidase; DNS-Az: dansyl azide; ECG: electrocardiogram; ESIPT: excited state intramolecular proton transfer; ETC: electron transfer chain; FBS: fetal bovine serum; FSN-AuNRs: fluoro-surfactant functionalized gold nanorods; GC: gas chromatography; GCE/GO@PQn: A 9,10-phenanthrene-quinone (PQn) tethered graphene oxide (GO) modified glassy carbon electrode; GC-MS: gas chromatography/mass spectrometry; H₂S: hydrogen sulfide; Hb: hemoglobin; HBO₂: hyperbaric oxygen; hCSFs: human corneal stromal fibroblasts; HPLC: high-performance liquid chromatography; HS-GC-MS: headspace-gas chromatography-mass spectrometry; LC: liquid chromatography; LC/MS: liquid chromatograph-mass spectrometer; LOD: detection limit; MAC: maximum allowable concentration; Mb: myoglobin; MB: methylene blue; MBB: methylene blue method; MOFs: metal-organic frameworks (Co/Fe-MOFs: catalytic MOFs); MTQ: 1-methyl-2-thioquinolone; N₂O: nitrous oxide; NaHS: sodium hydro-sulfide; NBDCl: 4-chloro-7-nitro-2,1,3-benzoxadiazole; NBDSh: 7-nitrobenzofurazan-4-thiol; N, N-dpd: N,N-dimethyl-p-phenylene diamine; PET: photoinduced electron transfer; pH: potential-of-hydrogen; RHP: P-nitrobenzyl-based hypoxia probe; RP-HPLC: reversed phase high-performance liquid chromatography; SDB: sulfide dibimane; STEL: short term exposure limit; TA: tannic acid; TICT: twisted intramolecular charge transfer; TMB: 3,3',5,5'-tetramethylbenzidine; TPA: tris-methyl pyridine amine; TPP-H₂S: the TP(two-photon) fluorescent probe 3-benzothiazol-7-(7-nitrobenzofurazan)-2H-chromen-2-one; TWA: time weighted average

ARTICLE HISTORY

Received 28 October 2022
Revised 1 January 2023
Accepted 4 January 2023

KEYWORDS

Hydrogen sulfide;
thiosulfate; analytic method;
poisoning; forensic
medicine

Table of contents

1. Introduction	743	2.3.1. Colorimetry	747
2. Hydrogen sulfide	743	2.3.2. Chromatography/mass-spectrography	749
2.1. Production of H ₂ S	743	2.3.3. Electrochemical method	749
2.1.1. H ₂ S in nature	743	2.3.4. Fluorescence-based assays	750
2.1.2. H ₂ S in the organism	743	2.4. H ₂ S detection in H ₂ S poisoning	750
2.1.3. H ₂ S donors	743	3. Metabolite—thiosulfate	750
2.2. Forensic aspect of H ₂ S poisoning	744	3.1. Thiosulfate in H ₂ S poisoning	750
2.2.1. Pathological changes and mechanism	744	3.2. Analytic aspects of thiosulfate	751
2.2.2. Eyes	746	3.2.1. Sampling and storage	751
2.2.3. Head and face	746	3.2.2. Detection methods—thiosulfate	751
2.2.4. Lung and airway	746	3.2.2.1. Chromatography and mass spectrometry	751
2.2.5. Brain	746	3.2.2.2. Others	751
2.2.6. Heart	747	3.3. Forensic aspect of thiosulfate detection in H ₂ S	
2.2.7. Others	747	poisoning	751
2.3. Analytic aspects of H ₂ S	747	3.3.1. Body fluid	751

3.3.1.1. Blood	751
3.3.1.2. Urine	752
3.3.1.3. Others	752
3.3.2. Parenchymal organs and tissues	752
4. Other in H ₂ S poisoning	752
5. Conclusion	753
Acknowledgments	753
Declaration of interest	753
References	753

1. Introduction

Hydrogen sulfide (H₂S) is a kind of gaseous substance. It exists everywhere. For one thing, H₂S can act as a gaseous signal molecule for organics to regulate normal physiological balance. For another thing, excessive H₂S leads to a series of pathological damages and ultimately death. The accidents caused by H₂S used to occur mostly in occupational exposure (Li and Moore 2008). Currently, suicide using H₂S also have been a novel way of dying, and the rate has continuous increase (Sams et al. 2013). Indeed, it is difficult for forensic experts to detect and discern H₂S-involved death by anatomy alone.

Currently, the scale of the scientific literature in medicine over many decades proves the scientists' keen interest in H₂S poisoning. The latest review of the literature was published in 2022 and the lethal H₂S poisoning case was reported in 2020 on PubMed. "H₂S poisoning" is a constant keyword in search of H₂S poisoning cases. However, the publication searched was covered in different fields. To restrict the publication in the field of forensic medicine, other related keywords can also be applied to the search process at the same time, such as "case report", "murder", "autopsy" and so on. In this paper, a large of screenings and readings are conducted and the aim is to provide a more comprehensive analysis of the related knowledge on the forensic aspect of H₂S and metabolite detection. It mainly includes the introduction and analytical methods of H₂S, pathology and molecular biology of H₂S poisoning, and even metabolites of H₂S poisoning. We hope this paper can be beneficial for H₂S poisoning identification in the future.

2. Hydrogen sulfide

2.1. Production of H₂S

2.1.1. H₂S in nature

H₂S, a gas with the characteristic odor of rotten eggs, was used to be well known for its toxicity (Reiffenstein et al. 1992). It is present in a natural process, especially in the sulfur springs, volcanic gases, swamps, undersea vents, and so on. Moreover, some social activities could also cause H₂S production such as petroleum processing or farming activities (Haouzi et al. 2020). Generally, it is hard for us to feel H₂S in daily life. This is due to little H₂S in the atmosphere. And less than 1.0 ppm of H₂S was released by a natural process (Aventaggiato et al. 2020). In China, H₂S poisoning ranked

second in occupational poisoning after carbon monoxide poisoning. To prevent and reduce toxic damage of H₂S, there have been many regulations about toxic gas whether in China or other countries. For example, the maximum allowable concentration (MAC) of H₂S in China is stipulated in the workplace. It has required that the MAC is 10 mg/m³. Furthermore, the UK used two indexes of time-weighted average (TWA) and short-term exposure limit (STEL) to restrict H₂S concentration in occupational settings, with the TWA being 5 ppm and STEL being 10 ppm (Costigan 2003). Besides, other countries and regions have similar requirements.

2.1.2. H₂S in the organism

H₂S is also produced and utilized in mammals. In 1996, a study by Abe and Kimura reported the physiological effects of H₂S in the brain (Abe and Kimura 1996). Subsequently, plenty of studies began to explore the action of H₂S in health or diseases. For example, it had been noted that H₂S plays a physiological role in vasodilation, pro- or anti-inflammation, neuroprotection, and regulating oxidative stress responses and cell death (Yu et al. 2014; Kar et al. 2019; Chen et al. 2020; Parfenova et al. 2020; Yin et al. 2020).

Endogenous H₂S synthesis occurs in almost all cells. It is internationally recognized that the production of endogenous H₂S mainly relies on two pathways including enzymatic and non-enzymatic reactions. Proteases cystathionine- β -synthase (CBS), cystathionine γ -lyase (CSE) and 3-mercaptopyruvate sulfurtransferase are essential for the enzymatic pathway (Renga 2011; Kimura et al. 2015). CBS mainly locates in brain and liver (Zhang et al. 2013; Bazhanov et al. 2017) whilst CSE mainly locates in the heart, lung, and eyes. The remaining protein 3MST expresses in the mitochondria and cytosol of cells in organs such as the brain (Kimura 2015). Compared with CBS and CSE which could catalyze L-cysteine into H₂S in the cytoplasm, 3MST catalyzes the synthesis of H₂S in the mitochondria via cooperating with cysteine aminotransferase (CAT) or D-amino acid oxidase (DAO) (Zhang et al. 2020).

Without stimulation, the concentration of H₂S is stable and the record can be seen in the table listed by Zhang's team (Zhang et al. 2020). To maintain the appropriate H₂S level in organisms, it is necessary to eliminate the excessive H₂S. The metabolism of H₂S is involved in many aspects. First, the liver is the main site of H₂S metabolism. H₂S can be catalyzed into metabolites such as thiosulfate, persulfide, sulfite, and sulfate by specific enzymes (Tokuda et al. 2012). In blood and tissues, H₂S can also react with hemoglobin (Hb) or myoglobin (Mb) to reach the goal of cleanness (Pietri et al. 2011). Additionally, H₂S can be methylated to form dimethyl sulfide in the cytoplasm of cells or it also can be exhaled through the respiratory tract (Yin et al. 2020).

2.1.3. H₂S donors

H₂S is an inorganic compound mainly composed of one sulfur ion and two hydrogen ions. It can dissolve in water in the form of gas and ion. When the temperature is 20 °C, 242 ml

water can absorb 1 g H₂S (Li and Moore 2008), of which gas H₂S accounts for one-third while ion HS⁻ is for two-thirds in water. It is dangerous for experimenters when they utilize H₂S gas in a narrow space. Therefore, the significance of H₂S donors is to reduce this danger.

Literature had exhibited a huge number of achievements which were the modulation of pathophysiological responses involved by H₂S. However, it still takes time to transform basic research into clinical application in the future. Sodium hydrosulfide (NaHS) as one kind of H₂S donor is most commonly used. It is an inorganic sulfide with a simple structure and can release H₂S rapidly in water (Caliendo et al. 2010), so it is widely used to study the biological properties of H₂S in various diseases of the endocrine system (Bitar et al. 2018), cardiovascular system (Yoo et al. 2015) and nervous system (Xu et al. 2018). However, it could cause a deep drop in blood pressure due to its rapid release of H₂S (Bitar et al. 2018) which may cause adverse consequences. Another important compound is GYY4137 which is superior to NaHS in the aspect of releasing H₂S. It could imitate the production of endogenous H₂S *in vivo* (Rose et al. 2015). Unlike NaHS and GYY4137, L-cysteine is also recognized as a kind of H₂S-releasing molecules, it can be absorbed *in vivo* and act as a substrate to participate in the H₂S synthesis (Blachier et al. 2019). In addition to these classical roles, other donors have been investigated by articles including Na₂S.

Currently, it is a challenge for people to exploit the H₂S-releasing drugs applied to clinical medicine. There are two points necessary for drug development. First, as noted above H₂S is a kind of toxic and causes severe consequences, so safety is the primary issue that needed to consider in drug application. Another point is the controlled release of H₂S. Some high polymer materials can combine with an H₂S donor and then prolong the release time. For example, the conjugation of H₂S donor SATOs with porous silica nanoparticles can induce the H₂S slow release and exert protection (Sun et al. 2017). Some synthetics are currently in clinical trials, and some have been already in use. A report showed that several H₂S-releasing derivatives were applied for the treatment of inflammatory bowel diseases, joint pain, and irritable bowel syndrome (Hosoki et al. 1997). All in all, it is becoming increasingly clear that H₂S has enormous potential in clinical medicine, but the development of H₂S reagents also comes with some side effects. Using H₂S to commit suicide also has become a trend (Sams et al. 2013). The suicide utilizes the principle that a sulfur compound can react with an acid-base to form lethal H₂S gas (Ruder et al. 2015), such as the combination of an acidic toilet bowl cleaner and bath salts. These chemicals are easily acquired in everyday life and make suicide feasible.

2.2. Forensic aspect of H₂S poisoning

2.2.1. Pathological changes and mechanism

H₂S poisoning mostly occurs in unpredictable accidents. And it is usually seen in confined spaces, especially workplaces such as downhole operation, washing the slurry tank, and so on. Recently, with the development of H₂S-related products,

the number of suicide cases had increased. The first reported case of suicide occurred in Japan (2007) (Bott and Dodd 2013), and then, this method of suicide was welcomed and widely spread because of the extremely rapid and painless death (Bott and Dodd 2013). People who commit suicide usually choose small and confined spaces to kill themselves. Jane D summarized whole cases from 2007 to 2010 in the United States (Reedy et al. 2011), and four-fifths of these events occurred in the car. Furthermore, other articles showed similar characteristics and the car became the preferred suicide location. Therefore, it is significant for us to emphasize and care more about H₂S poisoning.

Currently, it emphasizes manifestation of H₂S poisoning mainly depends on the H₂S concentration and exposure duration (Gerasimon et al. 2007). Prior utilized rats to mimic the H₂S poisoning and investigated the concentration-time interactions (Prior et al. 1988). It showed that clinical symptoms worsened following the increase in exposure time and H₂S concentration. At the same time, this article also reported there was no obvious difference between exposure times when extremely high concentrations lead to rapid death. A case report showed high concentration (about 500 ppm) can cause rapid unconscious ("knockdown" effect) in only a few breaths. In this situation, some victims can be still rescued if exposure is promptly ceased (Moretti et al. 2020).

Currently, the main therapeutic methods involve nitrite therapy (such as sodium nitrite) and hyperbaric oxygen (HBO₂) therapy (Asif and Exline 2012). Nitrite therapy is recognized as a special antidote and is more used in the early stage of poisoning, especially in the first 5–10 min of exposure for maximum effectiveness (Gerasimon et al. 2007). Compare with H₂S, it has the advantage of binding to the cytochrome oxidase aa3 system to inform the methemoglobin and then inhibits the effect of H₂S. Hyperbaric therapy can promote the peripheral tissues to use oxygen as far as possible and alleviate hypoxic injury. Moreover, other therapies such as the use of antibiotic therapy (Ventura Spagnolo et al. 2019) are based on specific symptoms.

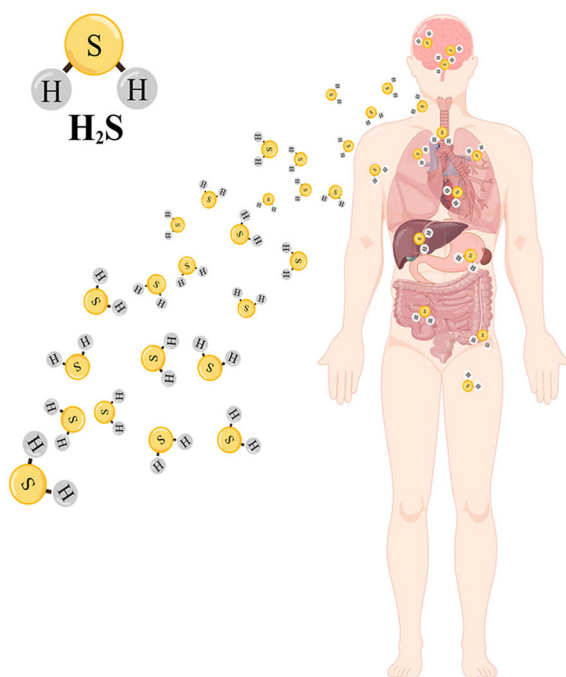
A review published by Haouzi's team in 2019 systematically showed the clinical symptoms after toxicosis including acute or long-term neurological effects (respiratory depression; coma; epilepsy and so on) (Haouzi et al. 2020). For mild patients, it doesn't have obvious symptom (Lindenmann et al. 2010). However, in most cases, the damage caused by H₂S poisoning is inevitable (Table 1).

In forensic medicine, we also summarize common pathological changes in different organs and tissues via analyzing H₂S-poisoning case reports (Figure 1).

Inhalation of a large amount of H₂S in a short time can cause intoxication and even death. It is recognized that H₂S poisoning depends on its binding to metalloproteins or cysteine residues of various proteins (Haouzi et al. 2016; Szabo 2018). It is generally accepted that H₂S prefers to combine with cytochrome C oxidase (CCO) which locates in mitochondria, thereby causing mitochondrial dysfunction and the failure of ATP formation. One experiment reported that H₂S exerts its toxicity via inducing a reductive shift in the NAD⁺/NADH redox couple of the electron transfer chain (ETC) in colon cells (Libiad et al. 2019). Another explained

Table 1. Clinical manifestation caused by H₂S of various concentrations.

H ₂ S airborne concentrations	Manifestation	References
10ppm (Within 60 min)	Ceiling concentration without irreversible harm	(Moretti et al. 2020)
<20ppm	odor perception—"rotten egg" scent	(Aventaggiato et al. 2020)
20–50ppm	Respiratory system: hyposmia; pulmonary; membrane irritation; Others: eye irritation;	(Bott and Dodd 2013; Aventaggiato et al. 2020)
50–100 ppm	Central nervous system: dizziness; Respiratory system: airway irritation Digestive system: nausea and vomiting; Others: keratoconjunctivitis and corneal ulceration;	(Moretti et al. 2020)
100–150 ppm	Central nervous system: unconsciousness (within a few minutes); Respiratory system: olfactory paralysis;	(Aventaggiato et al. 2020)
200–500 ppm	Central nervous system: headache; dizziness; disorientation; coma; Respiratory system: respiratory insufficiency; pulmonary edema; bronchospasm; rhinorrhoea; Digestive system: nausea; vomiting; diarrhea; Cardiovascular system: cyanosis; hemoptysis; palpitations; cardiac arrhythmia; tachycardia;hypotension; Others: photophobia; lacrimation;	(Gerasimon et al. 2007; Bott and Dodd 2013; Aventaggiato et al. 2020)
500–700ppm	Central nervous system: confusion; unconsciousness Cardiovascular system: hypotension Respiratory system: acute lung injury: bleeding and inflammation; respiratory arrest within 30–60 min; Collapse (knock down) Other: cyanosis	(Gerasimon et al. 2007; Tiwari 2008; Moretti et al. 2020)
700–1000 ppm	Central nervous system: asphyxial seizures; coma; unconsciousness Respiratory system: apnea Cardiovascular system: cardiopulmonary arrest; Collapse (knock down)	(Gerasimon et al. 2007; Bott and Dodd 2013; Aventaggiato et al. 2020; Moretti et al. 2020)
>1000 ppm	Immediate death after breath	(Bott and Dodd 2013)

**Brain**

greenish discoloration (sometimes);
cerebral edema and congestion.

Eyes

palpebral conjunctiva and ocular conjunctiva:
congestion or bleeding.

Head and Face

mucous membrane (mouth):
congestion and bleeding;

Lungs and airway

congestion;
trachea and bronchia: submucosal hemorrhage;
trachea and bronchia: red mucous and foamy fluid;
lung hyperinflation and hyperdistention;
pulmonary edema;
pulmonary hemorrhage.

Heart

congestion;
myocardial vacuolization.

Liver and spleen

congestion.

Pancreas

hemorrhage.

Kidneys and adrenal glands

congestion.

Skin

greenish discoloration and subcutaneous bleeding.

Figure 1. Common pathological changes of H₂S poisoning in forensic examination.

infusing the hyperbaric oxygen to reduce H_2S of CCO is useful (Lindenmann et al. 2010). Furthermore, some secondary mechanisms continued to be raised, for example, H_2S could break the disulfide bond of macromolecules and then exert toxicity (Wu et al. 2015). Next, we will describe the effects on different organs and some mechanisms of H_2S poisoning.

2.2.2. Eyes

H_2S can be produced in the retina and corneal layers in the bovine eye, and proteins CBS, CSE, and 3MST were detected in the eye tissues of humans, rats, and so on (Huang et al. 2017; Takır et al. 2021). Under physiological conditions, H_2S can regulate eye signal transduction and maintain normal eye activity (Njie-Mbye et al. 2012).

The eyes, one of the external organs, have the characteristic of being able to direct contact with air and have high sensitivity to irritation. Some reports claimed low concentration of H_2S could spark eye damage but not to other organs. Lambert et al. in 2006 published an article named "Hydrogen sulfide and sour gas effects on the eye". The issue made a summarization and analysis of clinical and pre-clinical data which was related to eye damage caused by H_2S (Lambert et al. 2006). Firstly, it introduced conjunctivitis, keratitis, photophobia, and even blindness was the result of poisoning. This article also highlighted the severities of damage that were concerned with concentration and time. For example, 20 ppm H_2S for 4 h caused eye damage, but 100 ppm for a short time would not. Therefore, it must need a description in detail of H_2S poisoning. Additionally, some special environments emitted air pollutants that may affect living quality and health, for example, people living in hog operations will have different levels of eye irritations (burning eyes and itching eyes) which had a positive association with the concentration of H_2S (Schinasi et al. 2011).

However, intrinsic molecular research on eye damage is currently very rare. One literature gave different concentrations of NaHS into human corneal stromal fibroblasts (hCSFs) to detect oxidative factors, inflammation, and cell proliferation-related indicators at different incubation times. It showed that H_2S exposure altered the expression of pro-inflammatory genes and anti-oxidative genes, therefore, it ultimately caused the fall in cell viability and proliferation (Balne et al. 2020).

2.2.3. Head and face

We have described the damages to the eye in detail above, which belongs to a dominating part of facial manifestations. Next, we will give some other abnormal occurrences of head and face.

It is common for forensic experts to find facial cyanosis in the deceased, especially in the lip and ears (Ago et al. 2008). High H_2S gas affects the physiological activity of cytochrome C oxidase and leads to respiratory depression, thereby inducing the accumulation of reduced hemoglobin in the body (Kiyota et al. 2002). This process promotes the formation of cyanosis. Generally, cyanosis is visible in some areas

consisting of the thin overlying epidermis and adequate blood vessels (McMullen and Patrick 2013).

Other features in people with H_2S poisoning also are recorded, such as facial congestion and so on. In one case, it showed white foam around the victim's nose and mouth, which indicated severe pulmonary edema (Knight and Presnell 2005). Apart from what we have described above, other records of facial damage are rare.

2.2.4. Lung and airway

Literature reported that H_2S enters the body through inhalation and exerts its toxic effects. Generally speaking, people can perceive H_2S based on the smell of "rotten eggs" (Knight and Presnell 2005). However, this warning was restricted. Olfactory nerve endings became gradually fatigued and even paralyzed with the increase of H_2S concentration (above 20 ppm) (Aventaggiato et al. 2020) which made people wrongly assume the gas had faded away. In this event, workers spent a lot of time staying in the workplace with H_2S , which resulted in non-cardiogenic pulmonary edema. When the concentration is high enough, the paralysis of olfactory nerve endings happened in a short time and "lighting" death would occur.

Next, we give some messages about the relationship between damage and H_2S concentration. It shows 50–100 ppm of H_2S in the air can cause airway irritation; 100–150 ppm can cause hyposmia, 200–300 ppm can cause respiratory insufficiency and pulmonary edema, and higher concentrations can cause suffocation and breathing paralysis. (Lambert et al. 2006; Aventaggiato et al. 2020; Mo et al. 2020; Moretti et al. 2020). In 1999, F. Almeida proposed that the lung was one of the main targets in peripheral organs (apnea precedes heart failure), and the vagus nerve-mediated lung-brain association was the main cause of asphyxia (Almeida and Guidotti 1999). In another study, Tanaka analyzed a case of three patients with H_2S poisoning. Pulmonary edema and bronchoscopy in the chest radiographs revealed different degrees of injured bronchi. One of them had obvious erythematous lesions which lasted for more than 10 days. Tanaka indicated their research showed that low concentration may lead to bronchial diseases (Tanaka et al. 1999). In the case of fatal H_2S poisoning, the autopsy results of four dairy farmers also showed characteristics of hyper-inflated and over-expanded lungs and pulmonary edema (Moretti et al. 2020). Unuma conducted animal experiments and then observed the congestion, inflammation and accumulation of mucopolysaccharides of the lung and bronchi. In this experiment, $TNF\alpha$ and $IL-1\beta$ were increased and MUC1 and 5B presented the same results by RT-PCR after poisoning. Furthermore, he used A594 lung epithelial cells to go on the simulated experiments and posed his view that H_2S exerts toxic effects in the lungs via the $NF-\kappa B-TNF\alpha/IL-1\beta-MUC1/5B$ pathway (Unuma et al. 2019).

2.2.5. Brain

Besides the lung, the brain is also recognized as one of the main targets (Gerasimon et al. 2007). Some survivors had

permanent neurological symptoms, which were closely related to the direct effect of H_2S on brain (Haouzi et al. 2020). In the case proposed by Gerasimon, the brain MRI with diffusion-weighted imaging showed the areas of restricted diffusion in the left superior cerebral hemisphere and the left basal ganglia and thalamus, and this image evidence further proved the direct brain injury (Gerasimon et al. 2007). Autopsy of some death cases sometimes showed green discoloration of the gray matter and substantia nigra, which were due to the combination of H_2S and blood-forming thiohemoglobin and thiomethylene hemoglobin (Smith et al. 2017; Ventura Spagnolo et al. 2019). It was worth noting that this phenomenon could also be seen in decomposed corpses because of the production of H_2S during the decomposing process. In addition, some nonspecific manifestations are also common in autopsy. Cerebral edema and congestion were often reported, and pathological examinations also showed neuronal hypoxic damage and pericapillary microhaemorrhages (Ventura Spagnolo et al. 2019).

2.2.6. Heart

Heart injury caused by cellular hypoxia and myocardial edema was widespread in the early stage of H_2S poisoning (Mo et al. 2020). The mechanism has been extensively studied. One theory is that H_2S can directly act on the mitochondria of cardiomyocytes and affect cellular respiration. The second is that H_2S has altered various ion channels before acting on mitochondria, including L-type Ca channels or depolarization-activated K^+ currents (Chen et al. 2016; Judenherc-Haouzi et al. 2016). The manifestation included dysrhythmias, angina attack, myocardial infarction, heart failure, and so on (Chen et al. 2016; Ventura Spagnolo et al. 2019). Clinical data displayed that the electrocardiogram (ECG) showed prominent massive ST-segment elevation which was following that of heart failure (Amino et al. 2009; Yang et al. 2018). Furthermore, some biochemical indices related to myocardial infarction also showed similar changes (Amino et al. 2009), but there was no obvious stenosis of the coronary arteries (Chen et al. 2016). Of which, yang also proposed the degree of ST-segment elevation correlated with the degree of poisoning, but I think it needs more evidence to verify.

H_2S would depress cardiomyocyte contractions by studying the isolated heart (Geng et al. 2004). A study performed on the heart of urethane-anesthetized rats showed that a transient hyperdynamic myocardial response may sustain for a short time, which meant sinus tachycardia and increased LV contractility. However, arterial blood pressure (BP) and heart rate (HR) decreased after a few seconds to a few minutes which meant the downregulation of the contractility (Judenherc-Haouzi et al. 2016; Kar et al. 2019). If the patient didn't get treatment on time, death would be coming. This is because 700 ppm H_2S would cause sudden cardiac arrest.

2.2.7. Others

Compared to the organs mentioned above, the description of others is lacking. It needs more time for specialists to

develop this field, and we can guess that it may be a key target of H_2S poisoning identification in the future.

2.3. Analytic aspects of H_2S

H_2S quantification in organs and tissues is usual for preclinical or clinical trials. Extensive clinical data had demonstrated stable H_2S is significant for a healthy person. For example, the data showed the H_2S concentration in the human brain ranges from 50 to 160 μM (Zhang et al. 2020), but it was downregulated in patients with Alzheimer's disease (Eto et al. 2002). Currently, multitudes of H_2S analytic methods have received study; hence we will introduce the main techniques used in biological samples, furthermore, some newly reported tests will contain in this review (Table 2).

2.3.1. Colorimetry

Colorimetry is the most traditional technique in H_2S detection. The principle is that the interaction of compounds can induce spectral and color changes.

The methylene blue method (MBB) is the classical colorimetric method (Ibrahim et al. 2021). It was first introduced by Fischer in 1883 (Quan and Lee 2021) and later modified by Fogo and his team (Fogo and Popowsky 1949). This principle mainly involves the formation of methylene blue (MB). Firstly, it added Zn^{2+} (ZnCl_2 , $\text{Zn}(\text{CH}_3\text{COO})_2$, or $\text{Zn}(\text{OAc})_2$) into the sample solution to form a zinc sulfide complex to inhibit the volatilization and oxidation of H_2S . Subsequently, the combination of an acidic environment and FeCl_3 could induce the release of H_2S in the zinc sulfide complex and promote the reaction of H_2S with N,N-dimethyl-p-phenylene Diamine (N,N-dpd) to form MB. The color change of MB can provide us with an intuitive impression, and MB also can be absorbed at 670 nm. It reported the absorbance and the color change are proportional to sulfide concentration. However, this method had some disadvantages of low sensitivity (the limit of detection (LOD) is about 10 μM), and difficult detection in complex samples. In order to overcome these weaknesses and make this method suitable for complex biological samples, other researchers offered more optimized ideas. For example, the existence of plasma protein would influence this experimental accuracy. Zheng chose to remove the related proteins by adjusting the potential-of-hydrogen (pH) and centrifuging before measuring the absorbance (Zheng et al. 2012). Currently, MBB is still one of the most widely used methods in biological samples, but other colorimetric methods also continued to be reported. We will take some simple examples.

In 2018, Kaushik et al. reported the colorimetric sensor 1, which consists of the phenolic group (Kaushik et al. 2018). This sensor can get deprotonated in the presence of $\text{H}_2\text{S}/\text{S}^{2-}$ and be detected by the UV-Vis absorption spectroscopy at 450 nm. To illustrate the effectiveness of this chemical compound, Kaushik did many related experiments such as the selection, sensitivity and time-dependent study. These experimental results showed sensor 1 had the advantages of and fast response time (within 1 min), and the detection limit was about 18.2 μM . Moreover, this sensor had used to measure

Table 2. Some analytic methods of H₂S used in biological samples.

No.	Main materials	Sample	Principle	LOD	Linear range
1 (Zheng et al. 2012)	MBB	Plasma (human)	In the presence of an acidic environment and FeCl ₃ , a zinc sulfide complex (the product of S ²⁻ and Zn ²⁺) can react with N, N-dpd to form the MB which can be absorbed in 670 nm.	1.6 μM	/
2 (Jarosz et al. 2013)	Microplate-based colorimetric method	Liver (mouse)	Free H ₂ S reacts with Ag ⁺ to promote the formation of Ag ₂ S nanoparticles, which have an obvious absorbance in the low-UV range.	8.70 μM	8.70–100 μM
3 (Liu and Feng 2014)	A visible light excitable colorimetric and fluorescent ESIPT probe	10%FBS	HS ⁻ mediated the thiolysis of compound which consisted of 4'-dimethylamino-3-hydroxyflavone and a dinitrophenyl group, and thereby exposing the OH group which can be absorbed at 538 nm.	0.1 μM	0–8 μM
4 (Zhang X et al. 2015)	FSN–AuNRs	Serum (Human and Mouse)	The interaction of gold and H ₂ S can induce the aggregation of FSN – AuNRs that can be detected at 518 nm and 648 nm by a UV–vis spectrophotometer.	0.2 μM	0.5–5 μM
5 (Zhao L et al. 2017)	Silver/dopamine nanoparticles	FBS and spiked serum samples	The combination of S ²⁻ and AgNPs would induce the formation of Ag ₂ S. This process made the absorbance of AgNPs at 400 nm decrease and the color of the solution change.	0.03 μM	2–15 μM
6 (Kaushik et al. 2018)	Azine based colorimetric sensor	Serum (Human and Mouse)	UV-Vis absorption spectrum detection of the developed H ₂ S/S ²⁻ -induced deprotonation at 450 nm	22.4 μM(mouse) 19.0 μM(human)	/
7 (Ma et al. 2020)	A colorimetric H ₂ S sensor (G4-Cu ²⁺)	Serum (Human)	S ²⁻ inhibits the formation of G4-Cu ²⁺ peroxidase mimetics, thereby influencing the oxidation of TMB (catalyzed by G4-Cu ²⁺). The color change is from blue to colorless and the detection of absorbance is at 652 nm wavelength.	7.5 nM	0.01–150 μM
8 (Zhao et al. 2021)	Iodide-enhanced Co/Fe-MOFs nanozyme	Serum (human)	The formation of blue oxTMB relied on the catalytic reaction of I ⁻ and Co/Fe-MOFs in the presence of H ₂ O ₂ . However, H ₂ S inhibits the formation of the oxTMB.	0.33 nM	1 nM–75 μM
9 (Vitvitsky and Banerjee 2015)	GC with Sulfur Chemiluminescence	Liver (mouse)	H ₂ S could be separated by GC and quantified by a 355 sulfur chemiluminescence detector.	0.5 pg	0–14 ng
10 (Varlet et al. 2015)	HS-GC-MS	Skeletal muscle (human)	HS-GC-MS with N ₂ O as an internal standard quantifies H ₂ S.	<1 μM	12.5–62.5 μM
11 (Roda et al. 2022)	HPLC-FLD based on monobromobimane derivatization	Serum (human)	S ²⁻ can react with MBB to form a chemical substance (SDB) that can be measured by HPLC-FLD.	0.5 μM	/
12 (Shalini Devi and Senthil Kumar 2018)	GCE/GO@PQn	Serum (human)	The GCE/GO@PQn modified electrode could selectively oxidize S ²⁻ by the proton-coupled electron-transfer reaction in neutral pH.	700 nM	1–100 μM
13 (Zhang et al. 2014)	A p-nitrobenzyl-based ratiometric fluorescent probe (RHP-2)	Brain (mouse)	The H ₂ S can react with RHP-2 relying on the nitro-reduction and induce the change in fluorescence.	270 nM	0–100 μM
14 (Chen et al. 2015)	A novel two-photon fluorescent probe	Brain, liver, and so on (mouse)	TPP-H ₂ S (no fluorescence) can react with HS ⁻ to induce the formation of 4-thio-7-nitrobenzofurazan and release the fluorophore.	0.12 μM	0.5–20 μM
15 (Li et al. 2015)	A reaction-based fluorescent switch-on sulfide sensor	Plasma, heart, and so on (rat)	The azo group can lock the rhodol fluorophore and show the nonfluorescent form, but S ²⁻ could reverse it and transfer the nonfluorescence to the high fluorescence.	500 nM	0–180 μM

(continued)

Table 2. Continued.

No.	Main materials	Sample	Principle	LOD	Linear range
16 (Alyan et al. 2021)	1, 5 Dansyl azide fluorescent probe (DNS-Az)	Plasma (human)	The probe DNS-Az can react with H ₂ S and form the fluorescent dansyl amide. The fluorescence can be detected by the in-house POCT portable spectrofluorometer	1 μM	3–300 μM

AgNPs: silver nanoparticles; DNS-Az: Dansyl Azide; ESIPT: excited state intramolecular proton transfer; FBS: fetal bovine serum; FSN-AuNRs: fluorosurfactant functionalized gold nanorods; GC: gas chromatography; GCE/GO@PQn: A 9,10-phenanthrenequinone (PQn) tethered graphene oxide (GO) modified glassy carbon electrode; H₂S: hydrogen sulfide; HPLC-FLD: high performance liquid chromatography with fluorescence detection; HS-GC-MS: headspace-gas chromatography-mass spectrometry; LOD: detection Limit; MBB: monobromobimane; MOFs: Metal-organic frameworks (Co/Fe-MOFs: catalytic MOFs); N, N-dpd: N,N-dimethyl-p-phenylene Diamine; pH: potential-of-hydrogen; RHP: p-nitrobenzyl-based hypoxia probe; RP-HPLC: reversed phase high-performance liquid chromatography; SDB: sulfide dibimane; TMB: 3,3',5,5'-tetramethylbenzidine; TPP-H₂S: the TP(Two-photon) fluorescent probe 3-benzothiazol-7-(7-nitrobenzofurazan)-2H-chromen-2-on

the H₂S concentration in human and mouse serum samples and got a good result.

In 2020, Ma suggested a novel colorimetric assay that was applied to serum detection (Ma et al. 2020). The detection was performed with an amplified signal based on G-quadruplex-Cu²⁺ (G4-Cu²⁺) peroxidase mimetics. The G4-Cu²⁺ complex had a high catalytic effect on the oxidation of 3,3',5,5'-tetramethylbenzidine (TMB) in the presence of H₂O₂ in the MES buffer. However, H₂S would have a strong reaction with Cu²⁺ and competitively inhibited G4-Cu²⁺ formation. TMB would turn from blue to colorless with the increasing H₂S, and it had a good linear relationship. To ensure practicability, the author verified various aspects, including the selective assay, recovery rate analysis (96.5–105.3% in the human serum), and so on. The detection limit was 7.5 nM, and the detection range was from 0.01 to 150 μM.

In addition, more colorimetric methods can be found in Table 2.

Compared with colorimetry, others such as chromatography, electrochemical method, and fluorescence-based assays were also appropriate for H₂S detection, which were introduced by Zhao in 2020 (Zhao et al. 2020).

2.3.2. Chromatography/mass-spectrography

Since the twenty first century, the development of chromatography technology has been approaching maturity. No matter gas chromatography (GC) or liquid chromatography (LC) (Ubuka et al. 2001; Lan et al. 2020), both of them are now widely reported in H₂S examination. GC is usually applied for H₂S quantification in coal, biogas and waste gas. Meanwhile, it can be used in the determination of biological samples.

Vitvitsky and Ruma (Vitvitsky and Banerjee 2015) introduced a GC with sulfur chemiluminescence method. The fundamental principle was that H₂S can be separated from other sulfur compounds to form sulfur monoxide by GC. And then, the product would react with ozone to form sulfur dioxide emitting light by a 355 sulfur chemiluminescence detector. As a result, the detected light intensity signal maintained a linear relationship with the sulfur content in the sample. According to statistics, this detection method could find sulfur at the pg level with a lower detection limit of 0.5 pg. Therefore, it was more sensitive than colorimetry. Moreover, Headspace-Gas Chromatography-Mass Spectrometry (HS-GC-MS) is another technology. It had been used in lethal

H₂S poisoning cases. Varlet (Varlet et al. 2015) introduced a modified method of HS-GC-MS. He used nitrous oxide (N₂O) as a novel internal standard and then measured the free H₂S level of muscle in the sealed headspace vials.

Shen et al. (2015) measured H₂S in biological samples using the monobromobimane method coupled with reversed-phase high-performance liquid chromatography (RP-HPLC). By combining the s-alkylation of two monobromobimane with H₂S, their derivatized products sulfide dibimane (SDB) could be formed at room temperature. The fluorescent SDB would be analyzed via RP-HPLC and the detection limit was 2 nM. Indeed, although this method may be effective, it had extremely high restrictions on sample preparation and reaction conditions in the process of the derivatization reaction. Considering the experimental analytical conditions (temperature, pH and so on) influencing the stability of the free H₂S, Roda et al. revised related steps based on this theory, and made the quantification more accurate (Roda et al. 2022).

2.3.3. Electrochemical method

The use of electrodes also promotes H₂S analysis. It has the potential of being rapid and convenient measurements. It usually divides into two kinds of devices: ion-selective electrodes and amperometric H₂S sensors (Zhao et al. 2020). At present, electrode reported is mainly used for the measurement of H₂S in the environment or in living animals. For example, Li et al. (2017) published an article about the design of ultrasensitive and facile electrochemical H₂S measurement based on a ligand exchange reaction (LOD: 5 nM). The core of this design was the formation of a Cu²⁺/Cys/Au electrode. The principle of H₂S quantification was the S²⁻ can competitively inhibit the link between cysteine and Au, resulting in a drop of Cu²⁺ from the electrode and creating a current which is consistent with H₂S concentration. This analytical method had been applied in rat brain microdialysate. However, whether it can be used in other biological samples (especially blood, homogenate tissue, and so on); still needs to be explored. In my opinion, its application has a high probability in body fluids.

In 2018, Shalini Devi reported a 9,10-phenanthrenequinone-tethered graphene oxide modified electrode (GCE/GO@PQn). The GCE/GO@PQn could selectively oxidize S²⁻ by the proton-coupled electron-transfer reaction in neutral pH and the linear range is 1–100 μM. It has been used to

examine the H_2S in human serum (Shalini Devi and Senthil Kumar 2018).

2.3.4. Fluorescence-based assays

At present, some research groups pay more attention to the design of fluorescence probes, thereby constructing new methods of H_2S measurement. With the high development of fluorescence imaging technology, a large number of novel probes have been reported over the years. In 2014, Zhang reported an p-nitrobenzyl-based ratiometric fluorescent probe (RHP-2), which is based on the principle of the nitro-reduction of H_2S , and it was successfully used in the mouse hippocampus. In 2015, Li et al. exhibited a novel fluorescent probe based on the azo group. In general, an azo group could induce the rhodol fluorophore to exist in the form of non-fluorescence (Li et al. 2015). However, this situation can be changed when sulfide occurs. A biocompatible reaction happened via interaction between the sulfide and the azo group and then caused the increasing fluorescence. The fluorescence intensity had a linear relationship with sulfide concentration and the low detection limit was 500 nM. This technology has been successfully applied to H_2S quantification in rat plasma and tissues such as the brain, lungs, kidneys, etc.

In 2021, Alyan utilized 1, 5-dansyl azide fluorescent probe to measure the H_2S concentration in human plasma (Linear range: 3–300 μM ; LOD: 1 μM). The mechanism is that the probe dansyl azide (DNS-Az) can react with H_2S and form the fluorescent dansyl amide, and the fluorescence can be detected by the in-house POCT portable spectrofluorometer.

Among these analytical methods, colorimetry has the advantages of convenience and rapid detection, but the defect is also obvious. For example, the primal MBB lacks selectivity and could be interfered with colored substances (Alyan et al. 2021). The methods of GC or MS have high sensitivity, but they need specialized skills to operate because of the complicated procedures and expensive equipment. Compared with other methods, the procedure of GC or MS is time-consuming (Jarosz et al. 2013). The detection of electrochemical method is also fast, but its sensitivity is lower than that of GC/MS, and most of them are used in the gas H_2S in the environment. At last, the development of fluorescence-based assays gets great progress, it is mainly used for living images and it has the characteristic of rapid response and noninvasive detection, but it also needs to improve their sensitivity and selection. Furthermore, we find most of these methods are used to detect the H_2S in simple biological samples, such as serum or fetal bovine serum (FBS). Whether they are appropriate for H_2S quantification in complicated biological samples, it is still a mystery. It is important and interesting for everyone to verify in the future.

2.4. H_2S detection in H_2S poisoning

H_2S is synthesized in the cytoplasm and mitochondria via the trans-sulfuration pathway at a steady rate (Kabil and Banerjee 2014). As a kind of gasotransmitter, H_2S usually can be dissolved into the blood at the pH of 7.35–7.45 and converted

into the state of gas (H_2S) and liquid (HS^- or S^{2-}), which is consistent with the situation in the water. The amount of H_2S in healthy subjects is still tough to define. Reference values provided by papers had obvious variations depending on the method of determination. For example, the results of the GC-MS and IC for H_2S concentration in plasma were from 35 to 80 μM and 100 to 800 nM, respectively. However, it is now certain that endogenous H_2S concentration in plasma is below micromolar (Zhang et al. 2020). Furthermore, unlike insulin, no studies have reported variation of H_2S in physiological conditions, so H_2S was recognized as a stable gas in a healthy person.

Unlike carbon monoxide (CO), there are seldom changes in the color of blood after H_2S poisoning and it isn't helpful for detection. It may be interesting in detecting H_2S levels. A study by Haouzi, a researcher who focuses on heart injury by H_2S , showed that the estimated concentration of gaseous H_2S in the arterial blood was about $33.26 \pm 6.43 \mu\text{M}$ after H_2S infusion when cardiac contractility decreased (Judenherc-Haouzi et al. 2016). And the same methods showed that H_2S was null in the baseline conditions. However, it is difficult in H_2S detection of the deceased in H_2S poisoning cases which depends on many factors. On one hand, a great many forensic reports indicated no H_2S was found in blood (Sastre et al. 2013). It attributed to the rapid degradation of inhaled H_2S via reaction with the ferrous structure of hemoglobin after entering the bloodstream. On the other hand, the process of corpse putrefaction can produce the gas H_2S and may hinder the exact identification in poisoning cases. A paper (Barbera et al. 2017) supported this opinion that the postmortem H_2S concentrations of fatal H_2S poisoning and control groups were 8.7–28.6 and 2.2–32.7 mg/L, respectively, and these data had small differentiation. Only the detection of H_2S concentration is difficult to analyze the cause of death. Besides quantitative analysis, the author suggested identification of H_2S poisoning must follow by additional pieces of evidence, such as histopathological changes in eye and airway irritation.

Nonetheless, we still found some case reports supporting H_2S as a direct shred of evidence. Sastre (Sastre et al. 2013) shared an article about the quantitation of H_2S -related death. A mother and her child died in the kitchen of their home. The dissection took place within 12 h and the samples (blood and lung) were preserved in the sealed headspace vials at $+4^\circ\text{C}$. After 72 h, H_2S analysis was performed by HS-GC-MS, the related value can be seen in Table 3. In this analysis process, the author also added other groups to guarantee that the values were not artificial or from corruption. Moreover, other literature also suggested that H_2S in the lungs and liver was more stable than that in the blood, which may give us new hope about H_2S quantitation of H_2S -related death.

3. Metabolite--thiosulfate

3.1. Thiosulfate in H_2S poisoning

As H_2S has the characteristic of volatilization and is easily oxidized (Quan and Lee 2021), the H_2S metabolism in tissues often happens in a short time, and thiosulfate is the main

by-product (Maebashi et al. 2011). In 2008, the researchers, M. Hildebrandt et al. explored the intrinsic mechanism of the conversion of sulfide to thiosulfate which occurred in mitochondria (Hildebrandt and Grieshaber 2008). Furthermore, other experiments analyzed the thiosulfate and H_2S variation when comparing the groups (the health or the H_2S poisoning). These results indicated that thiosulfate can be identified as the nonspecific marker of H_2S poisoning and it has been the secondary evidence in medicolegal expertise (Caliendo et al. 2010). However, it should be noted that thiosulfate measurement is invalid when facing a high concentration of H_2S . This is because “lighting” death occurs and H_2S fails to be oxidized in time.

3.2. Analytic aspects of thiosulfate

3.2.1. Sampling and storage

Thiosulfate, which is always used in preclinical, clinical, or forensic trial samples, is usually quantified in plasma and urine in bioanalytical practice. Several pieces of the literature showed its stability under different circumstances. Experiment by Jin (Jin et al. 2020) showed that thiosulfate in whole blood gradually decreased at 4°C , and approximately 13% were lost after 96 h. Subsequently, the author proposed a new method for delaying degradation—ultrafiltration. Ultrafiltration was used to reduce the macromolecules such as proteins, and the thiosulfate degradation was less than 10% in blood samples with ultrafiltration after 96 h using a liquid chromatograph-mass spectrometer (LC/MS). In addition to blood, the storage of thiosulfate in urine was proved to exist for weeks at 4°C and for months at -20°C (Cole and Evrovski 1997).

3.2.2. Detection methods—thiosulfate

3.2.2.1. Chromatography and mass spectrometry. The methods of chromatography, mass spectrometry, or their combinations are commonly considered as conventional thiosulfate measurements such as high-performance liquid chromatography (HPLC) (Shea and Howell 1984), gas chromatography/mass spectrometry (GC-MS) (Kage et al. 1991), LC-MS/MS (Jin et al. 2016) and others. The development of these technologies had facilitated the detection of thiosulfate in clinical and forensic diagnoses. Of which, GC/GC-MS was recognized as a standard method in the thiosulfate measurements. Shigetoshi Kage used the GC and GC-MS in the quantitative analysis of thiosulfate in the body fluids (Kage et al. 1991) and a sensitivity $0.003\text{ }\mu\text{mol/mL}$ is described.

Chwatko (Chwatko and Bald 2009) used the HPLC to disclose the concentration of thiosulfate in human urine. Firstly, 2-chloro-1-methylquinolinium tetrafluoroborate (CMQT) was used to precolumn derivatization before the separation of the reversed-phase liquid chromatography. In this process, CMQT could react with thiosulfate to form 1-methyl-2-thioquinolone (MTQ) in a water environment. Then, to detect the MTQ, ultraviolet detection technology was performed at analytical wavelengths of 375 nm which could bring an advantage in analysis. This result showed a linear relationship

between the thiosulfate and the low detection limit was $0.3\text{ }\mu\text{M}$.

Furthermore, the use of LC-MS/MS in the thiosulfate analysis was firstly reported by Jin (Jin et al. 2016). He analyzed the content of thiosulfate in the serum of the deceased (samples from H_2S poisoning or freezing to death) by improving the separation technology of LC to facilitate mass spectrometry measurement. Moreover, he added different concentrations of thiosulfate to serum in order to verify the recovery rate. As a result, the grand mean of the recovery rate was 99.53%. Therefore, it supported this method was appropriate for thiosulfate analysis.

3.2.2.2. Others. Compared to these chromatography and mass spectrometry, other methods are under development. So far, there are still some good references. Dong (Dong et al. 2017) in 2017 demonstrated a new strategy for efficient colorimetric detection of thiosulfate. The construction of sensor was based on the theory that the stabilizer tannic acid (TA) had good adsorption with AgNPs, thereby a redox reaction would rapidly occur in the presence of thiosulfate. Then, S^{2-} from this redox reaction had an ion exchange with Ag^+ and formed Ag_2S which attached to the surface of AgNPs. These series of processes ultimately led to an evident color change from yellow to brown which can be seen by the naked eye. In UV-vis spectroscopy analysis, the intensity of the UV-vis absorption had an obvious decrease as the color develops toward brown at 419 nm. Meanwhile, it also indicated that the thiosulfate concentration had a linear relationship with the absorption intensity with the low detection limit was $0.2\text{ }\mu\text{M}$.

Berthonnaud (Berthonnaud et al. 2021) analyzed the characterization of $[(\text{TMPA})\text{Cu}(\text{II})(\text{S}_2\text{O}_3)]$ in the state of solid or solution. The ligand of the Tris-Methyl Pyridine Amine (TMPA) named $\text{Cu}(\text{II})$ was used to involve the formation of $(\text{TMPA})\text{Cu}(\text{II})(\text{S}_2\text{O}_3)]$. Subsequently, its solution can be monitored via UV-Vis, EPR and cyclic voltammetry, while the solid was examined by X-ray crystallography. Both of these methods could detect the thiosulfate level.

3.3. Forensic aspect of thiosulfate detection in H_2S poisoning

3.3.1. Body fluid

3.3.1.1. Blood. Under normal circumstances, the amount of thiosulfate is extremely low, less than $0.003\text{ }\mu\text{mol/mL}$ (Ago et al. 2008). When H_2S is inhaled, it enters the blood circulation and is catalyzed to form thiosulfate which is more stable than H_2S in plasma. Literature showed the value of thiosulfate was the same at the time of 4 and 24 h after death. For the reason above, thiosulfate detection in the blood is regarded as the main basis for the diagnosis of H_2S poisoning (Maebashi et al. 2011; Maseda et al. 2017). Excessive thiosulfate can be found in most of the deceased blood in the situation of the H_2S hit. The thiosulfate in a lethal individual is from 0.025 to $0.143\text{ }\mu\text{mol/mL}$ (Bott and Dodd 2013). According to some forensic cases reported in the literature,

Table 3. H₂S detection by HS-GC-MS in a case of H₂S poisoning.

No.	Deceased	Age	Lung (mg/kg)	Blood	Manifestations	Sample storage
1(Sastre et al. 2013)	Mother	26	1.46	ND	Multiorgan congestion	+4 °C; Sealed storage in headspace vials
2(Sastre et al. 2013)	Child	<1	1.92	ND	① Multiorgan congestion; ② Lung and brain edema	

ND: not detected.

we made a conclusion. As shown in Table 4, about 32 people died from H₂S poisoning. Compared with the normal baseline level, the thiosulfate concentration in the blood was many times higher. This result provided strong evidence for H₂S poisoning identification. Moreover, the values of thiosulfate shown in this table had obvious differences. For one thing, it depends on the concentration of inhaled H₂S. For another, the different autopsy times may be a factor. Other factors include the time interval between poisoning and death, age, and so on.

However, it is risky to rely solely on thiosulfate testing for poisoning determination. Firstly, thiosulfate cannot be detected when the death happened within seconds in the situation of extremely high concentrations of H₂S. This is because it doesn't have enough time for H₂S metabolism to transform thiosulfate. Secondly, thiosulfate degradation caused by inappropriate blood sample preservation or a too long time of death had an influence, making the value below the limit of analytical methods. Then no thiosulfate can be detected. A paper (Jin et al. 2020) has shown that thiosulfate still would decrease in the process of proper preservation. The last reason is attributed to the sensitivity of the experimental methods. The difference in methods may make a distinction and choosing the wrong method will cause the wrong measurement result. However, it is worth noting that not all thiosulfate elevations in the blood are caused by H₂S. This reminds us to pay attention to discrimination.

3.3.1.2. Urine. Other than blood, urine is an important body fluid matrix. Whether in clinical or forensic medicine, urine testing is one of the commonly used methods.

Nowadays, no reports involved H₂S detection in urine because of its volatility. More attention is emphasized to the thiosulfate measurement. In general, thiosulfate in urine is below 0.03 µmol/ml in healthy subjects (Maseda et al. 2017). When exposed to large amounts of H₂S, the thiosulfate level in most situations may elevate a lot. In non-fatal H₂S poisoning cases, the survey of thiosulfate in the urine is valuable (Aventaggiato et al. 2020). However, the situation was reversed in the fatal H₂S poisoning cases. As shown in Table 5, the reference values of the deceased were exhibited. We could realize that undetectable thiosulfate was more common in urine than in blood samples. This is mainly attributable to the H₂S metabolism. For example, Moretti (Moretti et al. 2020) observed this phenomenon. Four workers died in the same accident, but only one victim had detected thiosulfate in urine and others weren't. This reminded us the detection of thiosulfate in urine is not stable for the identification of H₂S poisoning. In other words, once excessive thiosulfate is detected in urine, it must be given enough attention.

3.3.1.3. Others. Only one article reported other body fluids such as cerebrospinal fluid in H₂S poisoning cases, and the thiosulfate concentration in cerebrospinal fluid was 0.004–0.314 µmol/ml (Maebashi et al. 2011).

No reports involved thiosulfate detection in the vitreous humor. Nevertheless, forensic workers believe that the substance in the vitreous humor is relatively stable after death. Therefore, thiosulfate detection in the vitreous humor may provide evidence for forensic identification. This may be a breakthrough direction for H₂S poisoning identification in the future.

3.3.2. Parenchymal organs and tissues

Compared with blood and urine detection, thiosulfate of parenchymal organs was seldom reported in forensic cases. Moretti (Moretti et al. 2020) quantified the thiosulfate levels of different organs and tissues in four deceased who died of H₂S poisoning. The values of thiosulfate in the brain, lung, spleen, and muscle were 0–1.9, 2.8–5.4, 1.2–2.9 and 0–5.6 µg/g, respectively. And no thiosulfate was detected in fat. Another report by Lancia detected the thiosulfate concentration in the liver, lung and brain was 0.266, 1.013, and 1.111 mM/L, respectively (Lancia et al. 2013). In these data, we can find the thiosulfate concentration varies in the same organs. The inhaled H₂S concentrations were the main reason for these differences. In short, we advocated for thiosulfate detection in parenchymal organs and tissues which may promote further research about H₂S poisoning.

4. Other in H₂S poisoning

Unlike the above H₂S detection, the reported sulfide (S²⁻) level is also used to reflect H₂S concentrations *in vivo*. The sulfide concentration is controversial, with different reports are 0.96–3.2 µg/ml or below 0.05 µg/ml. These variations also still stem from differences in detection techniques.

In the paper, we conclude sulfide concentration in blood and urine of five H₂S-related cases (Table 6). Moreover, apart from the sulfide in blood and urine, sulfide in other organs was also reported using GC/MS by Moretti (Moretti et al. 2020). The researcher also evaluated the ratio of sulfide/thiosulfate. These results may reflect the circumstances of the deaths.

In 1989, a clear link between the brain sulfide level and H₂S-related mortality was reported. The analysis of gas dialysis indicated the brain level of sulfide was 1.57–2.30 µg/g under physiological conditions (Warenycia et al. 1989). Warenycia's team compared the sulfide concentration before and after H₂S poisoning in the brain (Warenycia et al. 1989). Their experimental data showed the sulfide in the brain was

Table 4. Blood thiosulfate concentration ($\mu\text{mol/ml}$) in H_2S -poisoning cases.

No.	Deceased	Thiosulfate ($\mu\text{mol/ml}$)	Analysis technique	Time interval (death to autopsy)
1(Zuka et al. 2012)	1	0.116	GC/MS	5h
2(Lancia et al. 2013)	1	1.02	GC/MS	48h
3(Kage et al. 1997)	2	0.025; 0.12	GC	24h
4(Ago et al. 2008)	2	Cardiac blood 0.089; 0.142	GC/MS	20h; 24 h
5(Aventaggiato et al. 2020)	2	Venous blood 0.040; 0.011 Arterial blood <0.0009	GC/MS	36h
6(Ventura Spagnolo et al. 2019)	3	0.19–0.32	GC/MS	24–28h
7(Kage et al. 2004)	4	0.11–0.23	GC/MS	24–26h
8(Moretti et al. 2020)	4	Cardiac blood 0.018–0.04 Femoral blood 0.018–0.02	GC/MS	4d
9(Maebashi et al. 2011)	13	0.014–0.648	GC/MS	7h–7d

Table 5. Urine thiosulfate concentration ($\mu\text{mol/ml}$) in H_2S -poisoning cases.

No.	Deceased	Thiosulfate ($\mu\text{mol/ml}$)	Analysis technique	Autopsy time
1(Zuka et al. 2012)	1	ca. 0.02	GC/MS	5h
2(Ventura Spagnolo et al. 2019)	2	0.09; 4.55	GC/MS	24–28h
3(Kage et al. 1997)	2	0.007; 0.95	GC	24h
4(Moretti et al. 2020)	4	<0.004–0.016	GC/MS	4d

Table 6. Sulfide concentration ($\mu\text{mol/ml}$) in H_2S -poisoning cases.

No.	Deceased	Blood ($\mu\text{g/ml}$)	Urine ($\mu\text{g/ml}$)	Analysis technique	Autopsy time
1(Zuka et al. 2012)	1	0.008	NA	GC/MS	5h
2(Ventura Spagnolo et al. 2019)	3	5.3–9.3	9.7/14.7 (2/3)	GC/MS	24–28h
3(Kage et al. 2004)	4	0.32–9.36	ND	GC/MS	24–26h
4(Moretti et al. 2020)	4	Cardiac blood 0.5–3 Femoral blood 0.5–1.2	<0.1	GC/MS	4d
5(Maebashi et al. 2011)	15	0.11–31.84 (15/15)	0.01–1.28 (10/15)	GC/MS	7h–7d

NA: not analysis; ND: not detected.

increased when H_2S was administered, and the sulfide level was positively correlated with H_2S concentration and lethality. Another study also reported that the sulfide in the brain was nearly $3.1 \mu\text{g/g}$ when given a 50% lethal dose of H_2S (Reiffenstein et al. 1992). Surprisingly, Warenycia further explored sulfide changes in different brain regions and cellular substructures. Therefore, the author believed that sulfide detected within 36 h post-death may play a significant role in the forensic identification of H_2S poisoning.

5. Conclusion

At present, the occurrence rate of H_2S poisoning is constantly increasing. However, there are still some problems with forensic identification. In our paper, we hope to provide a complete introduction to promote the development of identification.

Firstly, we shared the pathological changes and mechanisms of H_2S poisoning, it is very important in forensic medicine. Secondly, the analysis techniques of H_2S quantitation were listed. The goal is to provide new opinions about detecting H_2S poisoning. Finally, we introduced the application of thiosulfate (an H_2S metabolite as the nonspecific marker) in H_2S poisoning.

Above all, our article mainly serves for identification of H_2S poisoning. We also look forward to hearing the more development and application of the analytical methods in actual H_2S poisoning cases. Furthermore, our article has shown the exact amount of H_2S is still vague, the value

reported has a wide range and is influenced by the analytical methods. We also hope more specialists can join us and resolve this problem in the future, which may improve our understanding about H_2S .

Acknowledgments

The authors thank Figdraw (software for drawing figures). The authors also want to thank the anonymous reviewers selected by the editor. The reviewers gave us many valuable suggestions that guided us in revising the content of this article.

Declaration of interest

The authors report no conflict of interest. The manuscript was provided solely by the authors, and the authors' affiliations are as shown on the first page. Yiwu Zhou is a forensic specialist and is familiar with the research of forensic pathology and toxicology. He gives this article clear guidance. The other authors have made different contributions to this manuscript and all authors have been responsible for the content of this paper. There is no funding supporting the preparation of this work. In the past five years, the authors have never been involved in any legal or regulatory activities related this paper.

References

- Abe K, Kimura H. 1996. The possible role of hydrogen sulfide as an endogenous neuromodulator. *J Neurosci.* 16(3):1066–1071.
- Ago M, Ago K, Ogata M. 2008. Two fatalities by hydrogen sulfide poisoning: variation of pathological and toxicological findings. *Leg Med.* 10(3):148–152.

- Almeida AF, Guidotti TL. 1999. Differential sensitivity of lung and brain to sulfide exposure: a peripheral mechanism for apnea. *Toxicol Sci.* 50(2): 287–293.
- Alyan AK, Hanafi RS, Gad MZ. 2021. Point-of-care testing and optimization of sample treatment for fluorometric determination of hydrogen sulphide in plasma of cardiovascular patients. *J Adv Res.* 27:1–10.
- Amino M, Yoshioka K, Suzuki Y, Uemura S, Sakurai K, Fukushima T, Morita S, Otsuka H, Nakagawa Y, Yamamoto I, et al. 2009. Improvement in a patient suffering from cardiac injury due to severe hydrogen sulfide poisoning: a long-term examination of the process of recovery of heart failure by performing nuclear medicine study. *Intern Med.* 48(19):1745–1748.
- Asif MJ, Exline MC. 2012. Utilization of hyperbaric oxygen therapy and induced hypothermia after hydrogen sulfide exposure. *Respir Care.* 57(2):307–310.
- Aventaggiato L, Colucci AP, Strisciullo G, Favalli F, Gagliano-Candela R. 2020. Lethal hydrogen sulfide poisoning in open space: an atypical case of asphyxiation of two workers. *Forensic Sci Int.* 308:110122.
- Balne PK, Sinha NR, Hofmann AC, Martin LM, Mohan RR. 2020. Characterization of hydrogen sulfide toxicity to human corneal stromal fibroblasts. *Ann N Y Acad Sci.* 1480(1):207–218.
- Barbera N, Montana A, Indorato F, Arbouche N, Romano G. 2017. Evaluation of the role of toxicological data in discriminating between H₂S femoral blood concentration secondary to lethal poisoning and endogenous H₂S putrefactive production. *J Forensic Sci.* 62(2):392–394.
- Bazhanov N, Ansar M, Ivanciuc T, Garofalo RP, Casola A. 2017. Hydrogen sulfide: a novel player in airway development, pathophysiology of respiratory diseases, and antiviral defenses. *Am J Respir Cell Mol Biol.* 57(4):403–410.
- Berthonnaud L, Esmieu C, Mallet-Ladeira S, Hureau C. 2021. Solid-state and solution characterizations of [(TMPA)Cu(II)(SO₃)] and [(TMPA)Cu(II)(S₂O₃)] complexes: application to sulfite and thiosulfate fast detection. *J Inorg Biochem.* 225:111601.
- Bitar MS, Nader J, Al-Ali W, Al Madhoun A, Arefanian H, Al-Mulla F. 2018. Hydrogen sulfide donor NaHS improves metabolism and reduces muscle atrophy in type 2 diabetes: implication for understanding sarcopenic pathophysiology. *Oxid Med Cell Longev.* 2018:6825452.
- Blachier F, Beaumont M, Kim E. 2019. Cysteine-derived hydrogen sulfide and gut health: a matter of endogenous or bacterial origin. *Curr Opin Clin Nutr Metab Care.* 22(1):68–75.
- Bott E, Dodd M. 2013. Suicide by hydrogen sulfide inhalation. *Am J Forensic Med Pathol.* 34(1):23–25.
- Caliendo G, Cirino G, Santagada V, Wallace JL. 2010. Synthesis and biological effects of hydrogen sulfide (H₂S): development of H₂S-releasing drugs as pharmaceuticals. *J Med Chem.* 53(17):6275–6286.
- Chen J, Chen S, Mao W. 2016. A case of survival: myocardial infarction and ventricular arrhythmia induced by severe hydrogen sulfide poisoning. *Cardiology.* 135(1):43–47.
- Chen Q, Yang J, Li Y, Zheng J, Yang R. 2015. Sensitive and rapid detection of endogenous hydrogen sulfide by distributing in different mouse viscera via a two-photon fluorescent probe. *Anal Chim Acta.* 896: 128–136.
- Chen T, Tian M, Han Y. 2020. Hydrogen sulfide: a multi-tasking signal molecule in the regulation of oxidative stress responses. *J Exp Bot.* 71(10):2862–2869.
- Chwatko G, Bald E. 2009. Determination of thiosulfate in human urine by high performance liquid chromatography. *Talanta.* 79(2):229–234.
- Cole DE, Evroviski J. 1997. Quantitation of sulfate and thiosulfate in clinical samples by ion chromatography. *J Chromatogr A.* 789(1–2): 221–232.
- Costigan MG. 2003. Hydrogen sulfide: UK occupational exposure limits. *Occup Environ Med.* 60(4):308–312.
- Dong C, Wang Z, Zhang Y, Ma X, Iqbal MZ, Miao L, Zhou Z, Shen Z, Wu A. 2017. High-performance colorimetric detection of thiosulfate by using silver nanoparticles for smartphone-based analysis. *ACS Sens.* 2(8):1152–1159.
- Eto K, Asada T, Arima K, Makifuchi T, Kimura H. 2002. Brain hydrogen sulfide is severely decreased in Alzheimer's disease. *Biochem Biophys Res Commun.* 293(5):1485–1488.
- Fogo JK, Popowsky M. 1949. Spectrophotometric determination of hydrogen sulfide - methylene blue method. *Anal Chem.* 21(6):732–734.
- Geng B, Yang J, Qi Y, Zhao J, Pang Y, Du J, Tang C. 2004. H₂S generated by heart in rat and its effects on cardiac function. *Biochem Biophys Res Commun.* 313(2):362–368.
- Gerasimon G, Bennett S, Musser J, Rinard J. 2007. Acute hydrogen sulfide poisoning in a dairy farmer. *Clin Toxicol.* 45(4):420–423.
- Haouzi P, Sonobe T, Judenherc-Haouzi A. 2016. Developing effective countermeasures against acute hydrogen sulfide intoxication: challenges and limitations. *Ann N Y Acad Sci.* 1374(1):29–40.
- Haouzi P, Sonobe T, Judenherc-Haouzi A. 2020. Hydrogen sulfide intoxication induced brain injury and methylene blue. *Neurobiol Dis.* 133: 104474.
- Hildebrandt TM, Grieshaber MK. 2008. Three enzymatic activities catalyze the oxidation of sulfide to thiosulfate in mammalian and invertebrate mitochondria. *FEBS J.* 275(13):3352–3361.
- Hosoki R, Matsuki N, Kimura H. 1997. The possible role of hydrogen sulfide as an endogenous smooth muscle relaxant in synergy with nitric oxide. *Biochem Biophys Res Commun.* 237(3):527–531.
- Huang S, Huang P, Liu X, Lin Z, Wang J, Xu S, Guo L, Leung CK, Zhong Y. 2017. Relevant variations and neuroprotective effect of hydrogen sulfide in a rat glaucoma model. *Neuroscience.* 341:27–41.
- Ibrahim H, Serag A, Farag MA. 2021. Emerging analytical tools for the detection of the third gasotransmitter H₂S, a comprehensive review. *J Adv Res.* 27:137–153.
- Jarosz AP, Yep T, Mutus B. 2013. Microplate-based colorimetric detection of free hydrogen sulfide. *Anal Chem.* 85(7):3638–3643.
- Jin S, Hyodoh H, Matoba K, Feng F, Hayakawa A, Okuda K, Shimizu K, Haga S, Ozaki M, Terazawa K. 2016. Development for the measurement of serum thiosulfate using LC-MS/MS in forensic diagnosis of H₂S poisoning. *Leg Med.* 22:18–22.
- Jin S, Murakami M, Matoba K, Matoba T, Haga S, Ozaki M, Takeuchi A, Hyodoh H. 2020. Sample preparation method with ultrafiltration for whole blood thiosulfate measurement. *Leg Med.* 47:101765.
- Judenherc-Haouzi A, Zhang XQ, Sonobe T, Song J, Rannals MD, Wang J, Tubbs N, Cheung JY, Haouzi P. 2016. Methylene blue counteracts H₂S toxicity-induced cardiac depression by restoring L-type Ca channel activity. *Am J Physiol Regul Integr Comp Physiol.* 310(11):R1030–1044.
- Kabil O, Banerjee R. 2014. Enzymology of H₂S biogenesis, decay and signaling. *Antioxid Redox Signal.* 20(5):770–782.
- Kage S, Ikeda H, Ikeda N, Tsujita A, Kudo K. 2004. Fatal hydrogen sulfide poisoning at a dye works. *Leg Med.* 6(3):182–186.
- Kage S, Nagata T, Kudo K. 1991. Determination of thiosulfate in body fluids by GC and GC/MS. *J Anal Toxicol.* 15(3):148–150.
- Kage S, Takekawa K, Kurosaki K, Imamura T, Kudo K. 1997. The usefulness of thiosulfate as an indicator of hydrogen sulfide poisoning: three cases. *Int J Legal Med.* 110(4):220–222.
- Kar S, Kambis TN, Mishra PK. 2019. Hydrogen sulfide-mediated regulation of cell death signaling ameliorates adverse cardiac remodeling and diabetic cardiomyopathy. *Am J Physiol Heart Circ Physiol.* 316(6): H1237–H1252.
- Kaushik R, Ghosh A, Singh A, Jose DA. 2018. Colorimetric sensor for the detection of H₂S and its application in molecular half-subtractor. *Anal Chim Acta.* 1040:177–186.
- Kimura H. 2015. Signaling molecules: hydrogen sulfide and polysulfide. *Antioxid Redox Signal.* 22(5):362–376.
- Kimura Y, Toyofuku Y, Koike S, Shibuya N, Nagahara N, Lefer D, Ogasawara Y, Kimura H. 2015. Identification of H₂S₃ and H₂S produced by 3-mercaptopyruvate sulfurtransferase in the brain. *Sci Rep.* 5:14774.
- Kiyota K, Suzuki T, Adachi T, Nonaka A, Nishigawa Y, Hamabe Y. 2002. Fatal calcium polysulfide overdose presenting corrosive chemical injury of esophagus and sulf-hemoglobinemia. *Chudoku Kenkyu.* 15(2): 171–176.
- Knight LD, Presnell SE. 2005. Death by sewer gas: case report of a double fatality and review of the literature. *Am J Forensic Med Pathol.* 26(2): 181–185.
- Lambert TW, Goodwin VM, Stefani D, Stroscher L. 2006. Hydrogen sulfide (H₂S) and sour gas effects on the eye: A historical perspective. *Sci Total Environ.* 367(1):1–22.

- Lan LA, Wu SY, Meng XG, Jiang JJ, Zheng MY, Fan GR. 2020. A simple liquid chromatography tandem mass spectrometric method for fast detection of hydrogen sulfide based on thiolysis of 7-nitro-2, 1, 3-benzoxadiazole ether. *J Chromatogr A*. 1625:461243.
- Lancia M, Panata L, Tondi V, Carlini L, Bacci M, Rossi R. 2013. A fatal work-related poisoning by hydrogen sulfide: report on a case. *Am J Forensic Med Pathol*. 34(4):315–317.
- Li B, Li L, Wang K, Wang C, Zhang L, Liu K, Lin Y. 2017. Ultrasensitive and facile electrochemical detection of hydrogen sulfide in rat brain microdialysate based on competitive binding reaction. *Anal Bioanal Chem*. 409(4):1101–1107.
- Li L, Moore PK. 2008. Putative biological roles of hydrogen sulfide in health and disease: a breath of not so fresh air? *Trends Pharmacol Sci*. 29(2):84–90.
- Li X, Cheng J, Gong Y, Yang B, Hu Y. 2015. Mapping hydrogen sulfide in rats with a novel azo-based fluorescent probe. *Biosens Bioelectron*. 65: 302–306.
- Libiad M, Vitvitsky V, Bostelaar T, Bak DW, Lee HJ, Sakamoto N, Fearon E, Lyssiotis CA, Weerapana E, Banerjee R. 2019. Hydrogen sulfide perturbs mitochondrial bioenergetics and triggers metabolic reprogramming in colon cells. *J Biol Chem*. 294(32):12077–12090.
- Lindenmann J, Matzi V, Anegg U, Neuboeck N, Porubsky C, Fell B, Raber T, Ratzenhofer-Komenda B, Renner H, Klemen H, et al. 2010. Hyperbaric oxygen in the treatment of hydrogen sulphide intoxication. *Acta Anaesthesiol Scand*. 54(6):784–785.
- Ma Y, Xu M, Wang Y, Liu Z, Ye B. 2020. A highly sensitive and adjustable colorimetric assay of hydrogen sulfide by signal amplification based on G-quadruplex-Cu(2+) peroxidase mimetics. *Analyst*. 145(8):2995–3001.
- Maebashi K, Iwade K, Sakai K, Takatsu A, Fukui K, Aoyagi M, Ochiai E, Nagai T. 2011. Toxicological analysis of 17 autopsy cases of hydrogen sulfide poisoning resulting from the inhalation of intentionally generated hydrogen sulfide gas. *Forensic Sci Int*. 207(1–3):91–95.
- Maseda C, Hayakawa A, Okuda K, Asari M, Tanaka H, Yamada H, Jin S, Horioka K, Matoba K, Shiono H, et al. 2017. Liquid chromatography-tandem mass spectrometry method for the determination of thiosulfate in human blood and urine as an indicator of hydrogen sulfide poisoning. *Leg Med*. 24:67–74.
- McMullen SM, Patrick W. 2013. Cyanosis. *Am J Med*. 126(3):210–212.
- Mo W, Shen J, Huang X, Zhang Y, Zhang Z. 2020. Acute myocardial injury following hydrogen sulfide poisoning. *Toxicol Ind Health*. 36(10):750–758.
- Moretti M, Ballardini M, Siodambro C, Tronconi L, Osculati AMM, Freni F, Vignali C, Morini L. 2020. Fatal poisoning of four workers in a farm: distribution of hydrogen sulfide and thiosulfate in 10 different biological matrices. *Forensic Sci Int*. 316:110525.
- Njie-Mbye YF, Opere CA, Chitnis M, Ohia SE. 2012. Hydrogen sulfide: role in ion channel and transporter modulation in the eye. *Front Physiol*. 3:295.
- Parfenova H, Liu J, Hoover DT, Fedinec AL. 2020. Vasodilator effects of sulfuraphane in cerebral circulation: a critical role of endogenously produced hydrogen sulfide and arteriolar smooth muscle KATP and BK channels in the brain. *J Cereb Blood Flow Metab*. 40(10):1987–1996.
- Pietri R, Roman-Morales E, Lopez-Garriga J. 2011. Hydrogen sulfide and hemeproteins: knowledge and mysteries. *Antioxid Redox Signal*. 15(2): 393–404.
- Prior MG, Sharma AK, Yong S, Lopez A. 1988. Concentration-time interactions in hydrogen sulphide toxicity in rats. *Can J Vet Res*. 52(3):375–379.
- Quan FS, Lee GJ. 2021. Analytical methods for detection of gasotransmitter hydrogen sulfide released from live cells. *Biomed Res Int*. 2021: 5473965.
- Reedy SJ, Schwartz MD, Morgan BW. 2011. Suicide fads: frequency and characteristics of hydrogen sulfide suicides in the United States. *West J Emerg Med*. 12(3):300–304.
- Reiffenstein RJ, Hulbert WC, Roth SH. 1992. Toxicology of hydrogen sulfide. *Annu Rev Pharmacol Toxicol*. 32:109–134.
- Renga B. 2011. Hydrogen sulfide generation in mammals: the molecular biology of cystathionine-beta-synthase (CBS) and cystathionine-gamma-lyase (CSE). *Inflamm Allergy Drug Targets*. 10(2):85–91.
- Roda B, Zhang N, Gambari L, Grigolo B, Eller-Vainicher C, Gennari L, Zappi A, Giordani S, Marassi V, Zattoni A, et al. 2022. Optimization of a monobromobimane (MBB) derivatization and RP-HPLC-FLD detection method for sulfur species measurement in human serum after sulfur inhalation treatment. *Antioxidants (Basel)*. 11(5):939.
- Rose P, Dymock BW, Moore PK. 2015. GYY4137, a novel water-soluble, H₂S-releasing molecule. *Methods Enzymol*. 554:143–167.
- Ruder JB, Ward JG, Taylor S, Giles K, Higgins T, Haan JM. 2015. Hydrogen sulfide suicide: a new trend and threat to healthcare providers. *J Burn Care Res*. 36(2):e23–e25.
- Sams RN, Carver HW, 2nd, Catanese C, Gilson T. 2013. Suicide with hydrogen sulfide. *Am J Forensic Med Pathol*. 34(2):81–82.
- Sastre C, Baillif-Couniou V, Kintz P, Cirimele V, Bartoli C, Christia-Lotter MA, Piercecchi-Marti MD, Leonetti G, Pelissier-Alicot AL. 2013. Fatal accidental hydrogen sulfide poisoning: a domestic case. *J Forensic Sci*. 58(Suppl 1):S280–S284.
- Schinasi L, Horton RA, Guidry VT, Wing S, Marshall SW, Morland KB. 2011. Air pollution, lung function, and physical symptoms in communities near concentrated Swine feeding operations. *Epidemiology*. 22(2):208–215.
- Shalini Devi KS, Senthil Kumar A. 2018. A blood-serum sulfide selective electrochemical sensor based on a 9,10-phenanthrenequinone-tethered graphene oxide modified electrode. *Analyst*. 143(13):3114–3123.
- Shea M, Howell S. 1984. High-performance liquid chromatographic measurement of exogenous thiosulfate in urine and plasma. *Anal Biochem*. 140(2):589–594.
- Shen X, Kolluru GK, Yuan S, Kevil CG. 2015. Measurement of H₂S *in vivo* and *in vitro* by the monobromobimane method. *Methods Enzymol*. 554:31–45.
- Smith SS, Cannon DL, Fagan K, Weis CP. 2017. Occupational hydrogen sulfide fatalities and thiosulfate levels. *Am J Forensic Med Pathol*. 38(1):47–48.
- Sun X, Wang W, Dai J, Jin S, Huang J, Guo C, Wang C, Pang L, Wang Y. 2017. A long-term and slow-releasing hydrogen sulfide donor protects against myocardial ischemia/reperfusion injury. *Sci Rep*. 7(1):3541.
- Szabo C. 2018. A timeline of hydrogen sulfide (H₂S) research: from environmental toxin to biological mediator. *Biochem Pharmacol*. 149:5–19.
- Takur S, Semiz AT, Uydeş Doğan BS. 2021. Hydrogen sulfide is synthesized endogenously in both retinal artery and retina mostly via CSE. *Exp Eye Res*. 204:108443.
- Tanaka S, Fujimoto S, Tamagaki Y, Wakayama K, Shimada K, Yoshikawa J. 1999. Bronchial injury and pulmonary edema caused by hydrogen sulfide poisoning. *Am J Emerg Med*. 17(4):427–429.
- Tiwari RR. 2008. Occupational health hazards in sewage and sanitary workers. *Indian J Occup Environ Med*. 12(3):112–115.
- Tokuda K, Kida K, Marutani E, Crimi E, Bougaki M, Khatri A, Kimura H, Ichinose F. 2012. Inhaled hydrogen sulfide prevents endotoxin-induced systemic inflammation and improves survival by altering sulfide metabolism in mice. *Antioxid Redox Signal*. 17(1):11–21.
- Ubuka T, Abe T, Kajikawa R, Morino K. 2001. Determination of hydrogen sulfide and acid-labile sulfur in animal tissues by gas chromatography and ion chromatography. *J Chromatogr B Biomed Sci Appl*. 757(1): 31–37.
- Unuma K, Aki T, Yamashita A, Yoshikawa A, Uemura K. 2019. Hydrogen sulfide donor NaHS causes bronchitis with enhanced respiratory secretion in rats. *J Toxicol Sci*. 44(2):107–112.
- Varlet V, Giuliani N, Palmieri C, Maujean G, Augsburg M. 2015. Hydrogen sulfide measurement by headspace-gas chromatography-mass spectrometry (HS-GC-MS): application to gaseous samples and gas dissolved in muscle. *J Anal Toxicol*. 39(1):52–57.
- Ventura Spagnolo E, Romano G, Zuccarello P, Laudani A, Mondello C, Argo A, Zerbo S, Barbera N. 2019. Toxicological investigations in a fatal and non-fatal accident due to hydrogen sulphide (H₂S) poisoning. *Forensic Sci Int*. 300:e4–e8.
- Vitvitsky V, Banerjee R. 2015. H₂S analysis in biological samples using gas chromatography with sulfur chemiluminescence detection. *Methods Enzymol*. 554:111–123.

- Warenycia MW, Goodwin LR, Benishin CG, Reiffenstein RJ, Francom DM, Taylor JD, Dieken FP. 1989. Acute hydrogen sulfide poisoning. Demonstration of selective uptake of sulfide by the brainstem by measurement of brain sulfide levels. *Biochem Pharmacol.* 38(6):973–981.
- Warenycia MW, Reiffenstein RJ, Goodwin LR, Dieken FP. 1989. Brain sulfide levels in anaesthesia: a comparison with hydrogen sulfide intoxication. *Toxicol Lett.* 47(3):221–224.
- Wu SH, Chen KF, Chan YL. 2015. Severe hydrogen sulfide intoxication treated with therapeutic hypothermia. *Am J Emerg Med.* 33(9):1327 e1321–1322.
- Xu K, Wu F, Xu K, Li Z, Wei X, Lu Q, Jiang T, Wu F, Xu X, Xiao J, et al. 2018. NaHS restores mitochondrial function and inhibits autophagy by activating the PI3K/Akt/mTOR signalling pathway to improve functional recovery after traumatic brain injury. *Chem Biol Interact.* 286: 96–105.
- Yang Y, Hu D, Peng D. 2018. Diffuse ST-segment elevation after hydrogen sulfide intoxication. *J Emerg Med.* 54(2):241–243.
- Yin L, Gao S, Li C. 2020. Exogenous hydrogen sulfide alleviates surgery-induced neuroinflammatory cognitive impairment in adult mice by inhibiting NO signaling. *BMC Anesthesiol.* 20(1):12.
- Yoo D, Jupiter RC, Pankey EA, Reddy VG, Edward JA, Swan KW, Peak TC, Mostany R, Kadowitz PJ. 2015. Analysis of cardiovascular responses to the H₂S donors Na₂S and NaHS in the rat. *Am J Physiol Heart Circ Physiol.* 309(4):H605–H614.
- Yu YP, Chi XL, Liu LJ. 2014. A hypothesis: hydrogen sulfide might be neuroprotective against subarachnoid hemorrhage induced brain injury. *ScientificWorldJournal.* 2014:432318.
- Zhang J, Shan H, Tao L, Zhang M. 2020. Biological effects of hydrogen sulfide and its protective role in intracerebral hemorrhage. *J Mol Neurosci.* 70(12):2020–2030.
- Zhang L, Meng WQ, Lu L, Xue YS, Li C, Zou F, Liu Y, Zhao J. 2014. Selective detection of endogenous H₂S in living cells and the mouse hippocampus using a ratiometric fluorescent probe. *Sci Rep.* 4:5870.
- Zhang M, Shan H, Wang T, Liu W, Wang Y, Wang L, Zhang L, Chang P, Dong W, Chen X, et al. 2013. Dynamic change of hydrogen sulfide after traumatic brain injury and its effect in mice. *Neurochem Res.* 38(4):714–725.
- Zhao D, Zhang J, Zhou M, Zhou H, Gotor C, Romero LC, Shen J, Yuan X, Xie Y. 2020. Current approaches for detection of hydrogen sulfide and persulfidation in biological systems. *Plant Physiol Biochem.* 155: 367–373.
- Zhao XL, Liu JL, Xie FT, Yang T, Hu R, Yang YH. 2021. Iodide-enhanced Co/Fe-MOFs nanozyme for sensitively colorimetric detection of H₂S. *Spectrochim Acta A Mol Biomol Spectrosc.* 262:120117.
- Zheng Y, Liao F, Du JB, Tang CS, Xu GH, Geng B. 2012. Modified methylene blue method for measurement of hydrogen sulfide level in plasma. *Sheng Li Xue Bao.* 64(6):681–686.
- Zuka M, Chinaka S, Matsumoto Y, Takayama N, Hitomi Y, Nakamura H, Ohshima T. 2012. Fatal and non-fatal cases of lime sulfide exposure and pathogenetic mechanisms underlying pancreatic injury: case reports with an animal experiment. *J Forensic Leg Med.* 19(6): 358–362.

Copyright of Critical Reviews in Toxicology is the property of Taylor & Francis Ltd and its content may not be copied or emailed to multiple sites or posted to a listserv without the copyright holder's express written permission. However, users may print, download, or email articles for individual use.